GABARITO

QUESTÃO 1: $Y = A + BX + CX^2$

$$3 = A + B + C$$
$$3 = A + 2B + 4C$$
$$5 = A + 3B + 9C$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & 9 & 5 \end{array}\right] \begin{array}{l} L_2 \to L_2 - L_1 \\ L_3 \to L_3 - L_1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 3 & 0 \\ 0 & 2 & 8 & 2 \end{array}\right] L_3 \to L_3 - 2L_2 \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 2 & 2 \end{array}\right]$$

$$A + B + C = 3 \Rightarrow A = 5$$
$$B + 3C = 0 \Rightarrow B = -3$$
$$2C = 2 \Rightarrow C = 1$$

$$\boxed{Y = 5 - 3X + X^2}$$

QUESTÃO 2:

$$\det \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & -2 & 0 \\ -2 & 3 & -2 & 3 \\ 0 & -3 & 3 & 3 \end{bmatrix} L_3 \to L_3 + 2L_1 = \det \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & -2 & 0 \\ 0 & 3 & -2 & 9 \\ 0 & -3 & 3 & 3 \end{bmatrix}$$

$$= 1(-1)^{1+1} \det \begin{bmatrix} 1 & -2 & 0 \\ 3 & -2 & 9 \\ -3 & 3 & 3 \end{bmatrix} \begin{array}{l} L_2 \to L_2 - 3L_1 \\ L_3 \to L_3 + 3L_1 \end{array} = \det \begin{bmatrix} 1 & -2 & 0 \\ 0 & 4 & 9 \\ 0 & -3 & 3 \end{bmatrix}$$

$$= 1(-1)^{1+1} \det \begin{bmatrix} 4 & 9 \\ -3 & 3 \end{bmatrix} = 12 + 27 = \boxed{39}$$

QUESTÃO 3:

$$x = \frac{\begin{vmatrix} 5 & 8 \\ 4 & 9 \end{vmatrix}}{\begin{vmatrix} 7 & 8 \\ 6 & 9 \end{vmatrix}} = \frac{45-32}{63-48} = \frac{13}{15}$$

$$y = \frac{\begin{vmatrix} 7 & 5 \\ 6 & 4 \end{vmatrix}}{\begin{vmatrix} 7 & 8 \\ 6 & 9 \end{vmatrix}} = \frac{28-30}{63-48} = \frac{-2}{15}$$

QUESTÃO 4:

$$\left[\begin{array}{ccc|ccc} 4 & 3 & 2 & 1 & 0 & 0 \\ 5 & 6 & 3 & 0 & 1 & 0 \\ 3 & 5 & 2 & 0 & 0 & 1 \end{array}\right] L_1 \to L_1 - L_3 \quad \left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & -1 \\ 5 & 6 & 3 & 0 & 1 & 0 \\ 3 & 5 & 2 & 0 & 0 & 1 \end{array}\right] \begin{matrix} L_2 \to L_2 - 5L_1 \\ L_3 \to L_3 - 3L_1 \end{matrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & -1 \\ 0 & 16 & 3 & -5 & 1 & 5 \\ 0 & 11 & 2 & -3 & 0 & 4 \end{array}\right] L_2 \to L_2 - L_3 \quad \left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & -1 \\ 0 & 5 & 1 & -2 & 1 & 1 \\ 0 & 11 & 2 & -3 & 0 & 4 \end{array}\right] L_3 \to L_3 - 2L_2$$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & -1 \\ 0 & 5 & 1 & -2 & 1 & 1 \\ 0 & 1 & 0 & 1 & -2 & 2 \end{array}\right] L_2 \leftrightarrow L_3 \quad \left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 & -2 & 2 \\ 0 & 5 & 1 & -2 & 1 & 1 \end{array}\right] \begin{matrix} L_1 \to L_1 + 2L_2 \\ \\ L_3 \to L_3 - 5L_2 \end{matrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -4 & 3 \\ 0 & 1 & 0 & 1 & -2 & 2 \\ 0 & 0 & 1 & -7 & 11 & -9 \end{array}\right] \qquad A^{-1} = \begin{bmatrix} 3 & -4 & 3 \\ 1 & -2 & 2 \\ -7 & 11 & -9 \end{bmatrix}$$

QUESTÃO 5:

$$A^{-1} = \begin{bmatrix} 3 & -4 & 3 \\ 1 & -2 & 2 \\ -7 & 11 & -9 \end{bmatrix}$$

$$M = \begin{bmatrix} 3 & -4 & 3 \\ 1 & -2 & 2 \\ -7 & 11 & -9 \end{bmatrix} \begin{bmatrix} 125 & 152 & 60 \\ 229 & 237 & 116 \\ 181 & 169 & 95 \end{bmatrix} = \begin{bmatrix} 2 & 15 & 1 \\ 29 & 16 & 18 \\ 15 & 22 & 1 \end{bmatrix}$$

2, 15, 1, 29, 16, 18, 15, 22, 1

B O A # P R O V A